

OpenCPU SDK

SNTP API Notes

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This document is specifically for **N706B** and **N520** modules.

This document is intended for system engineers (SEs), development engineers, and test engineers.

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About This Document

Scope

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Audience

This document is intended for [system engineers \(SEs\)](#), [development engineers](#), and [test engineers](#).

Change History

Issue	Date	Change	Changed By
1.0	2023-05	Initial release	Gao Zheng

Conventions

Symbol	Indication
	This warning symbol means danger. You are in a situation that could cause fatal device damage or even bodily damage.
	Means reader be careful. In this situation, you might perform an action that could result in module or product damages.
	Means note or tips for readers to use the module

1 Overview

This document introduces the SNTP-related APIs in the OpenCPU SDK, specifically for obtaining network time using SNTP.

2 Data Structure

2.1 nwy_sntp_param_t

```
typedef struct
{
    int cid;
    char url[NWY_SNTP_URL_MAX_LEN];
    int timeout;
    int retry_times;
    char tz[NWY_SNTP_TZ_MAX_LEN];
    unsigned char dst;
}nwy_sntp_param_t;
```

2.2 nwy_sntp_result_type

```
typedef struct{
    NWY_SNTP_EVENT_E event;           /* event id */
    nwy_time_t update_time;
}nwy_sntp_result_type;

typedef enum
{
    NWY_SNTP_SUCCESS = 0,
    NWY_SNTP_FAIL,
    NWY_SNTP_DNS_ERROR,
    NWY_SNTP_NET_ERROR,
    NWY_SNTP_TIMEOUT
}NWY_SNTP_EVENT_E;
```

3 API

The interface function definitions are located in `nwy_sntp_api.h`.

3.1 `nwy_sntp_get_time`

Format	<code>nwy_error_e nwy_sntp_get_time(nwy_sntp_param_t *sntp_param, nwy_sntp_resultcb cb_func);</code>
Description	Fetches network time via SNTP.
Parameters	<code>sntp_param</code> : Includes details such as CID, URL, and timeout. <code>cb_func</code> : Callback function for reporting the execution result.
Return Value	Success: <code>NWY_SUCCESS</code> Failure: other error codes